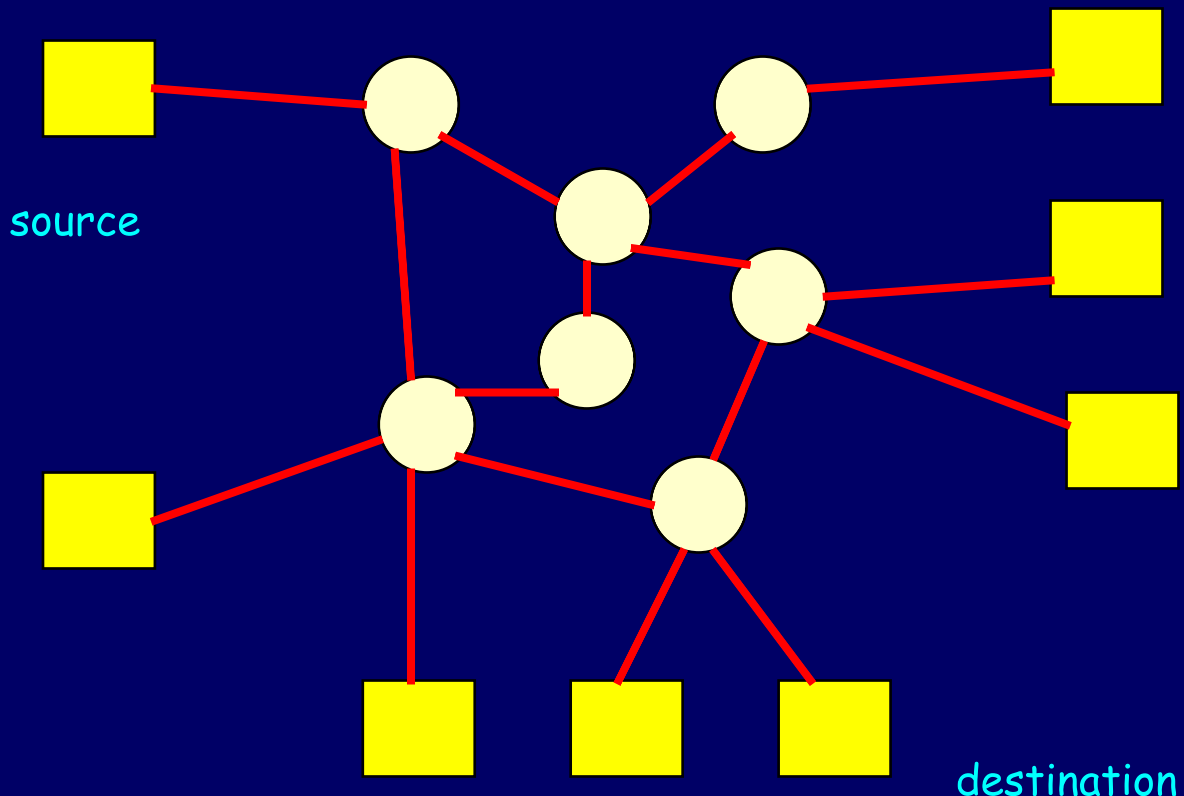
CprE 458/558: Real-Time Systems

Real-Time Networks --

Wide-Area Networks (WAN)

Real-Time communications: Introduction



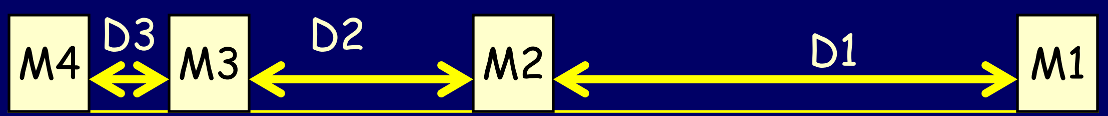
Performance metrics

- Bandwidth of the connection
- End-to-end delay: The total time the packet experienced from source to destination
- Delay jitter: It is the maximum variation in delay experienced by packets that travel across the connection
- Packet Loss: Percentage of packets lost
- The nature of the applications dictate the kind of performance requirements required

Performance metrics

- Delay

- Delay jitter



- Delay-jitter = Max_delay – Min_delay
  - In the example, Delay-jitter = (D1 – D3)

Applications and Guarantee requirements

- Interactive applications require
  - Bound on both delay and delay-jitter
  - Can tolerate occasional message loss
  - Examples: continuous media traffic (video or audio playback)

- Discrete applications require
  - Error-free service
  - Can tolerate both delay and jitter
  - Examples: File transfer, Image retrieval

Providing performance guarantees: Issues

- Choice of the packet scheduling algorithm at the intermediate node (switch)

- The message scheduling algorithms at the switches determine the order in which the packets from different connections are serviced

Approaches to Real-time Communication

- Pure circuit switching: It reserves the entire physical/virtual channel for the connection. E.g., telephone networks

- Pure packet switching: It can efficiently utilize network bandwidth but cannot provide real-time guarantees. E.g., Internet

- Packet-oriented switching: A virtual channel is established before transmission begins, employs statistical multiplexing to utilize bandwidth efficiently. E.g., ATM (Asynchronous Transfer Mode) network

Types of service

- Guaranteed service: (Deterministic or hard guaranteed service). This approach is conservative in resource reservation (for peak workload) and is the simplest method for real-time services.

- Predictive service: This service is meant for adaptive applications that can tolerate occasional violation of delay bound. Multimedia playback applications function well with this category of service.

- As-soon-as-possible service: This is best-effort service with priorities, the highest to be given to interactive burst traffic and the lowest to asynchronous bulk transfer. This category of service provides no guarantees, and no resources are reserved for it.

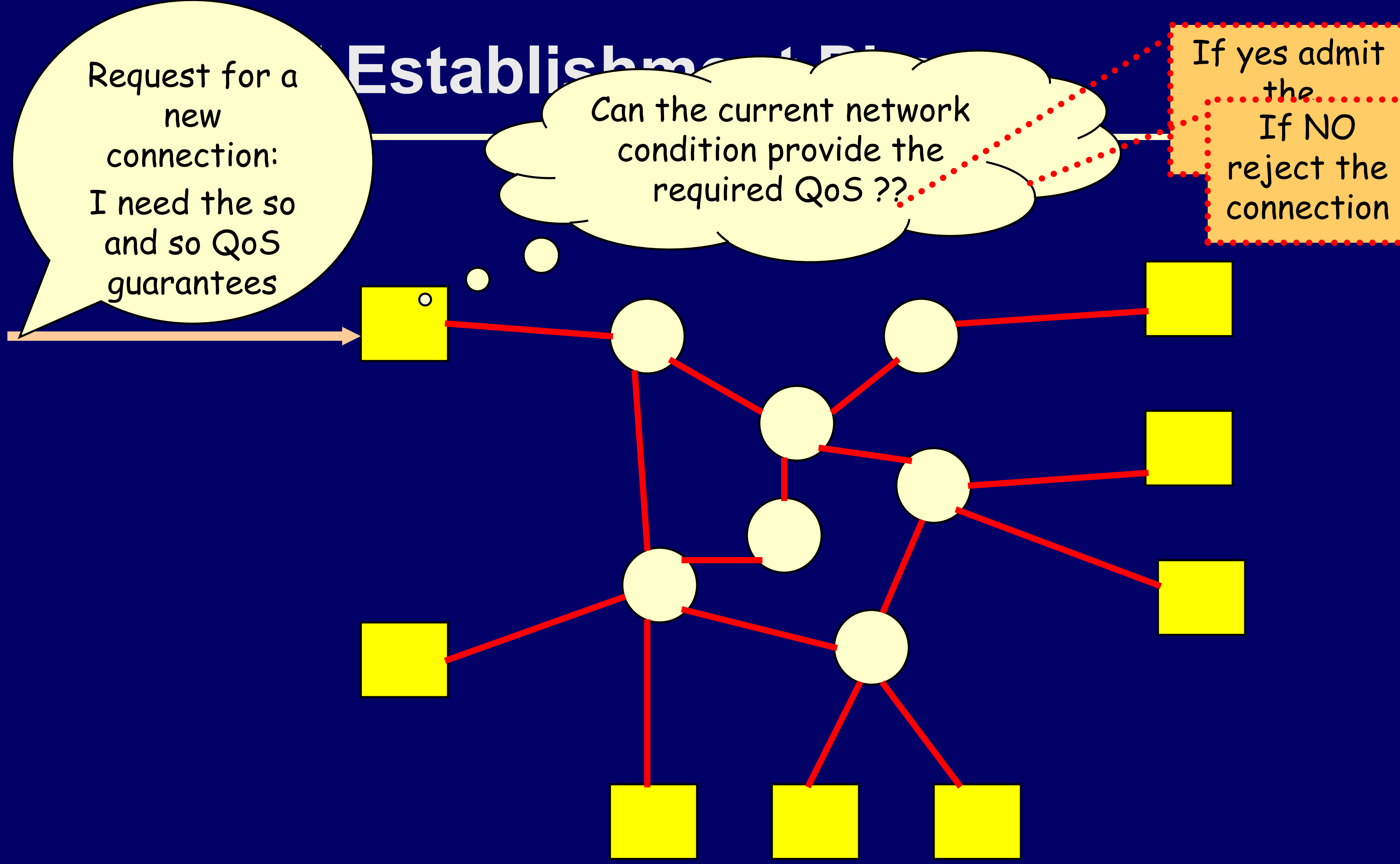
Real-Time Channel

- A virtual circuit that provides the required end-to-end QoS guarantees.
- QoS parameters: bandwidth, delay, delay jitter, packet loss, etc.

Life-cycle of a Real-Time Channel

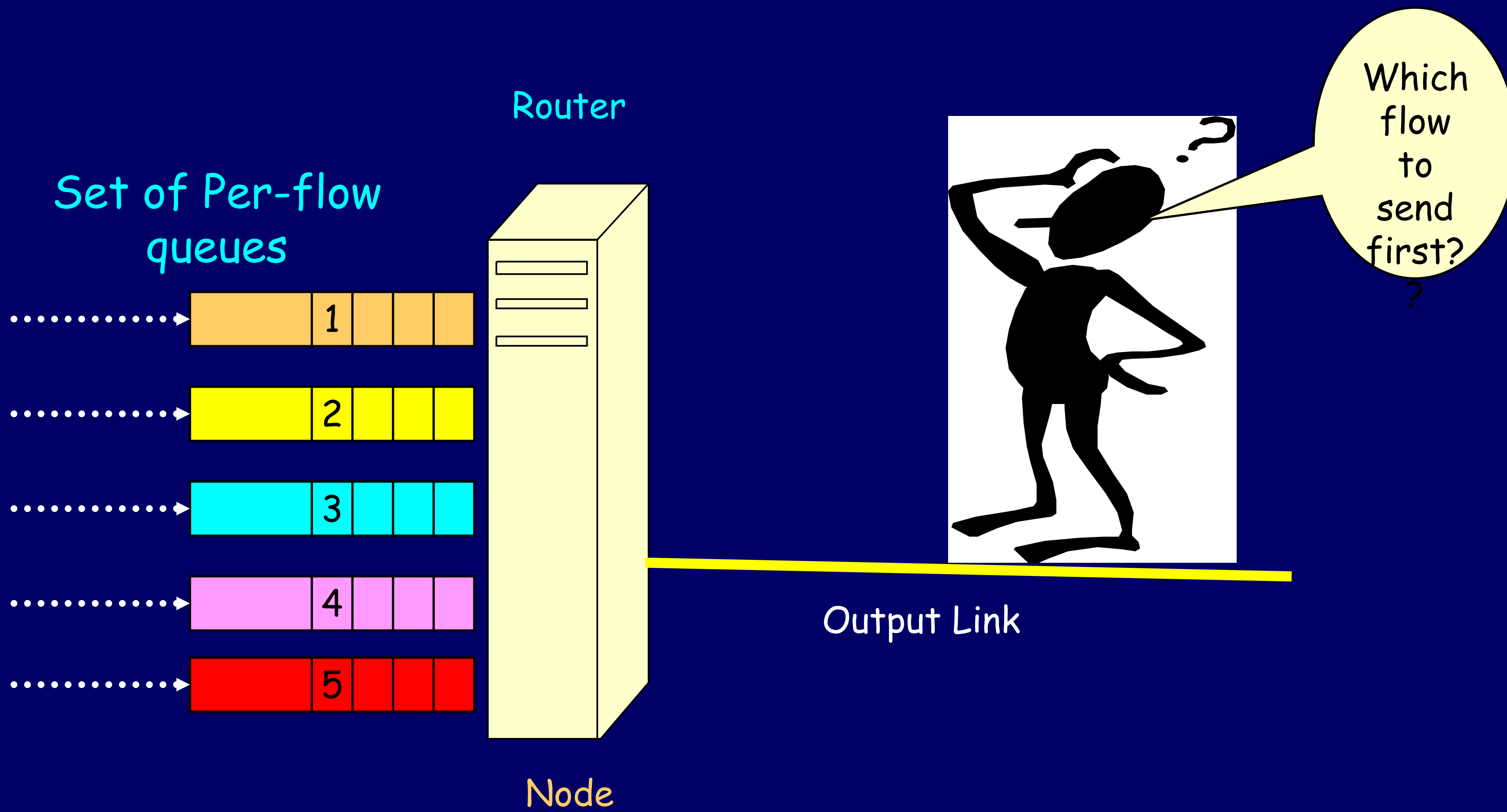
- Channel establishment phase
  - QoS routing
  - Resource reservation
- Data Transmission phase
  - Traffic policing/shaping
  - Packet scheduling
  - Rate adaptation
- Channel tear-down phase
  - Releasing session resources

Establishm



Request for a new connection:
I need the so and so QoS guarantees
Can the current network condition provide the required QoS ??
If yes admit the
If NO reject the connection

Run-time scheduling phase



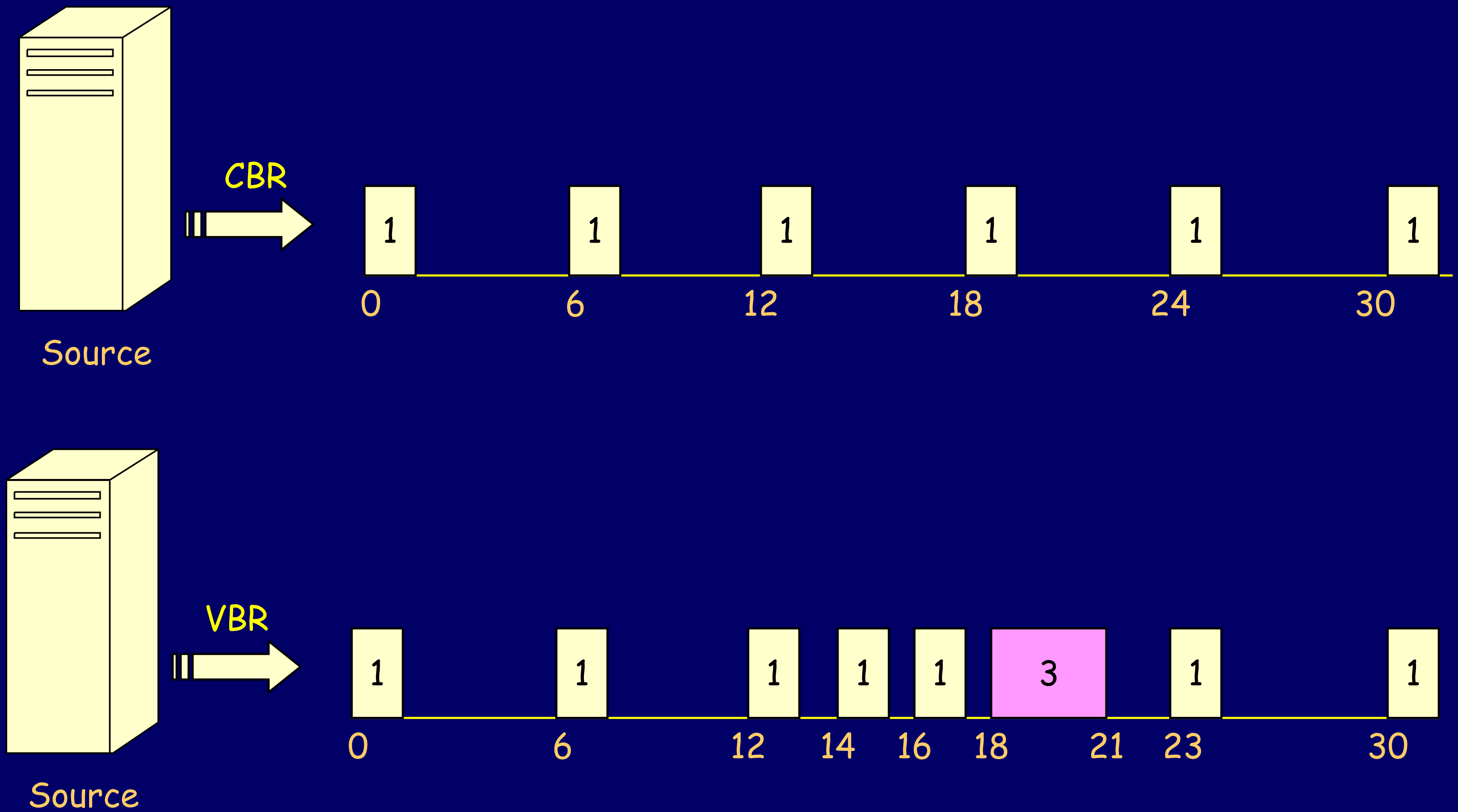
Characterization of Real-Time Traffic

- The traffic generated by the real-time sources fall in one of the two categories:

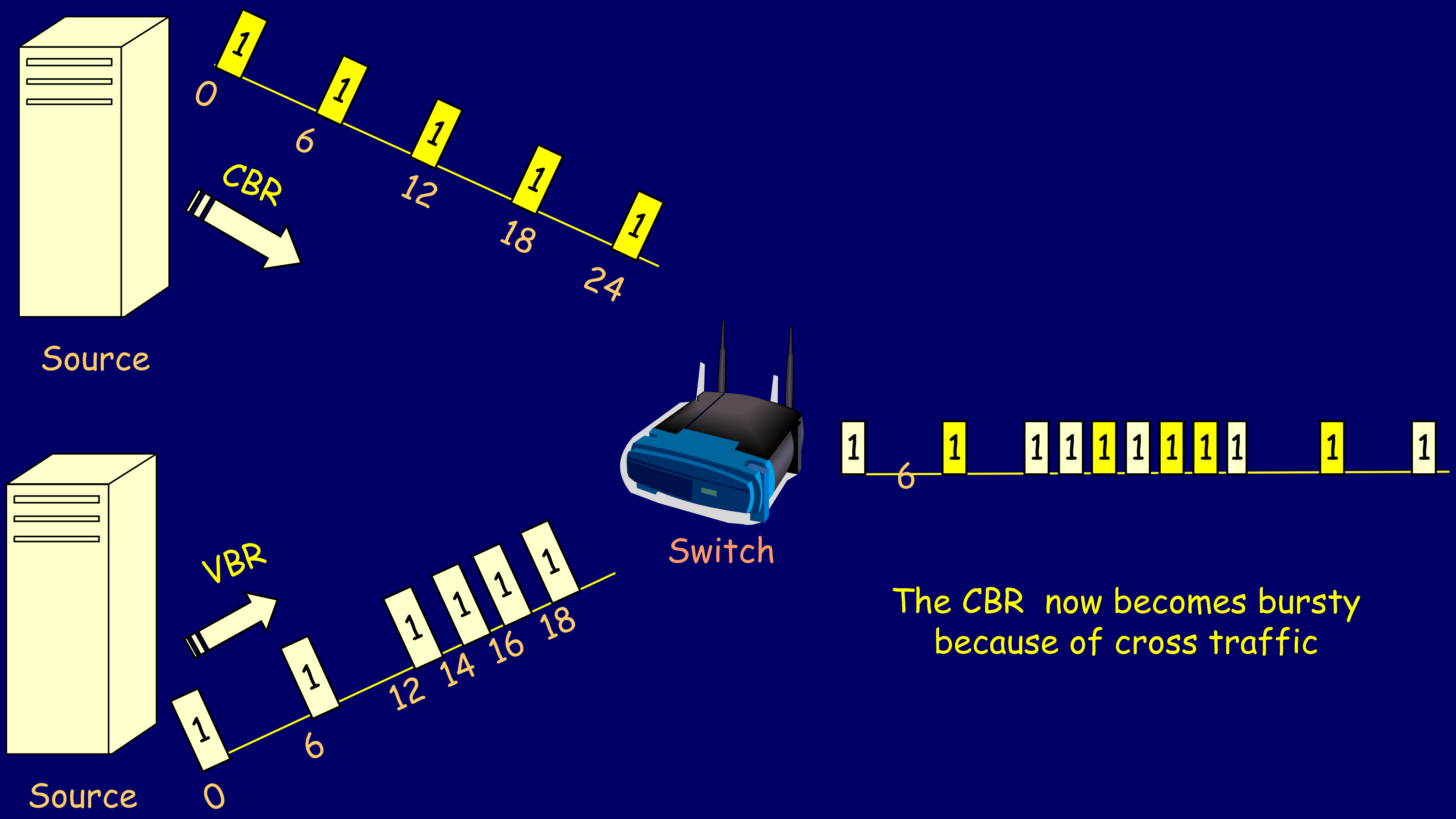
- Constant bit rate (CBR): In CBR, fixed-size packets are generated at regular intervals. It is smooth and nonbursty. The data generated by sensors (periodic).

- Variable bit rate (VBR): (1) fixed sized packets arriving at irregular intervals or (2) variable-sized packets arriving at regular intervals
  - Voice traffic (talk spurts alternate with periods of silence)
  - video source (different compression ratios result in variable size packets generated at regular intervals)

CBR and VBR examples



Change in Traffic characteristics



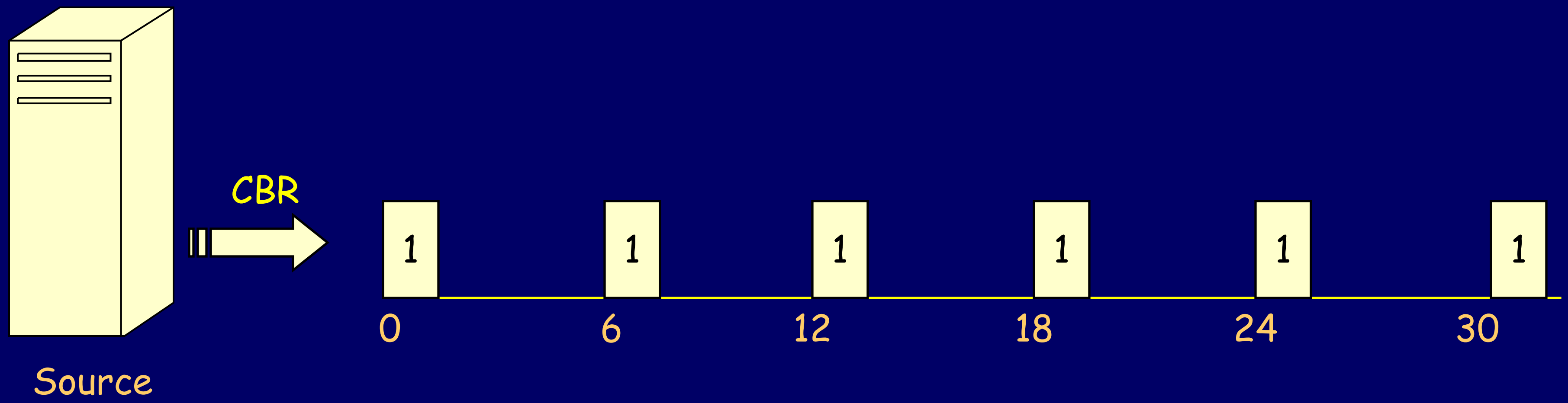
Traffic Models

- Peak-Rate Model: Most hard real-time systems use the peak-rate model for traffic characterization. The parameters of this model, for a connection i, are
  - Minimum inter arrival time between messages (T_i)
  - Maximum message rate ($1 / T_i$)
  - Maximum message length (μ_i)
  - End-to-end delay bound (D_i)

- The peak bandwidth requirement of the connection is (μ_i / T_i)

- The peak-rate model is exact only for the CBR traffic and overstates the bandwidth requirement for all VBR sources

Peak-rate model: Illustrative example



Minimum inter-arrival time (Ti) = 6 sec

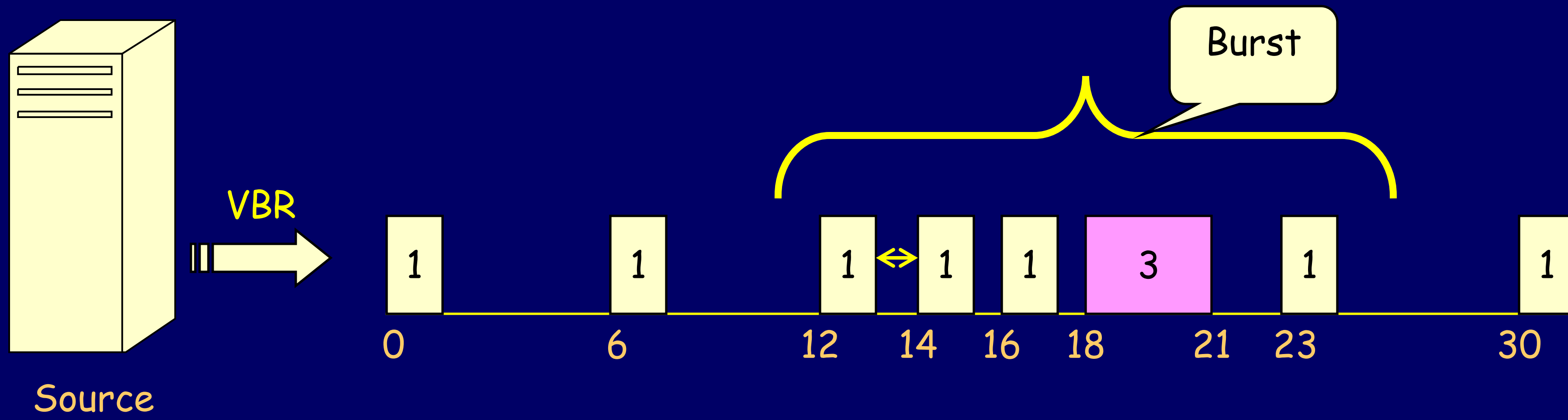
Maximum message rate (1 / Ti) = 1 / 6 = 0.16 message/sec

Maximum message length (μi) = 1 kbit

Bandwidth required = 1 / 6 = 0.16 kbits/sec

Exact B/W requirement

Peak-rate model: Illustrative example



Minimum inter arrival time (Ti) = 2 sec

Maximum message rate (1 / Ti) = 0.5 messages/sec

Maximum message length (μi) = 3 Kbits

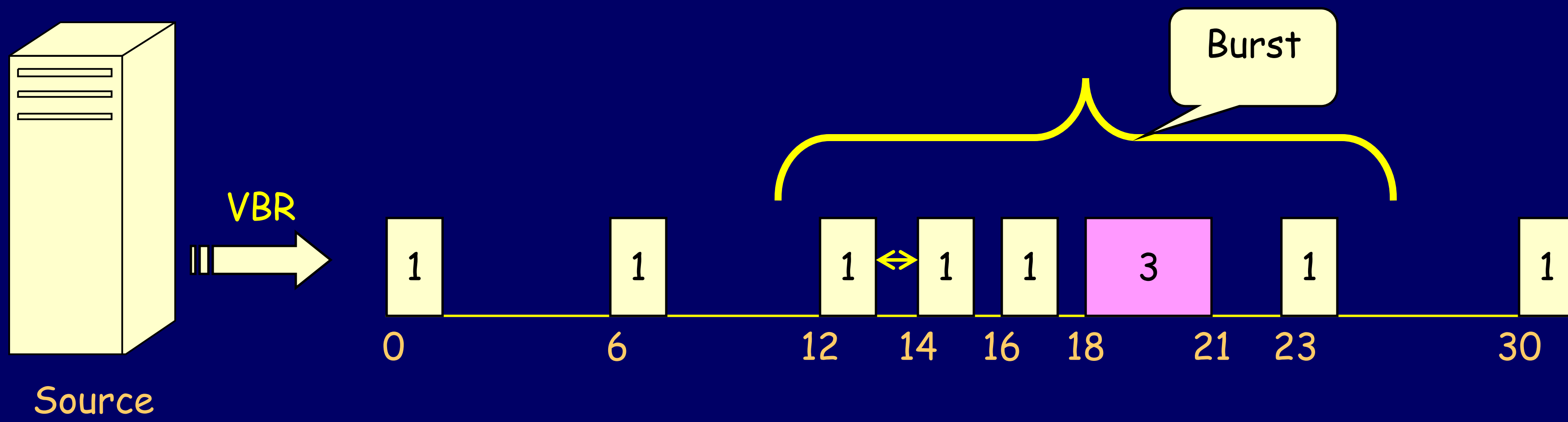
Peak bandwidth required = 3/2 = 1.5 Kbits/sec

An overstatement of the B/W requirement

Traffic Models (contd.)

- Linear Bounded Arrival Process (LBAP) Model
  - This model uses an additional parameter representing the maximum burst size (B_i)
  - In this model, the number of bits transmitted during any interval of length t is bounded by $B_i + (t / T_i)\ \mu_i$
  - This model can guarantee deterministic delay bounds

LBAP model: Illustrative example



Average inter-arrival time (Ti) = 6 sec

Maximum message rate (1 / Ti) = 0.16 messages/sec

Burst size (Bi) = 1 Kbits

Maximum message length (μi) = 3 Kbits

Bandwidth required = 3/6 + 1 = 1.5 Kbits/sec

An over estimate of the B/W req., but better than peak model